NPEC × BUAS

From root images to robot motion

Autonomous root tip targeting with computer vision,
PI control and reinforcement learning

Arabidopsis thaliana • HADES / NPEC



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The biological problem

Joint initiative of Wageningen University & Research and Utrecht University



CLIENT

Netherlands Plant Eco-phenotyping Centre (NPEC)

STUDY SPECIES

Arabidopsis thaliana

RESEARCH CONTEXT

Plant–microbe interactions

The opportunity

More consistent root inoculation could support larger experiments comparing responses across plant genotypes.

Sources: npec.nl/about-npec · dataset specimen imaging

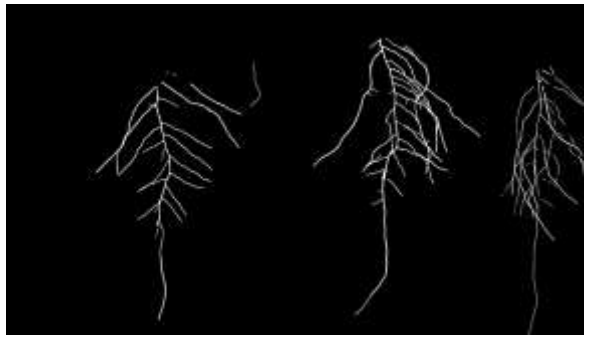
One pipeline, three disciplines

Computer vision provides the target; robotics delivers the motion



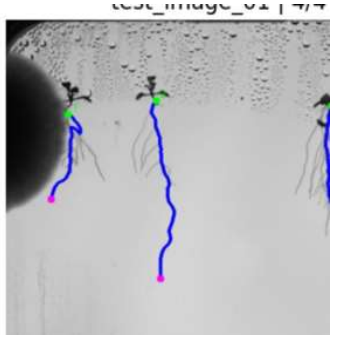
01 **HADES image**

Raw Petri dish



02 **Root segmentation**

512 px U-Net mask



03 **Root extraction**

Tip & length estimates

04 **Image coordinates**

05 **Assumed affine map**

06 **OT-2: PI / SAC**

Five plants, individually segmented

INPUT · MATURE SPECIMEN



OUTPUT · FIVE INSTANCE LABELS

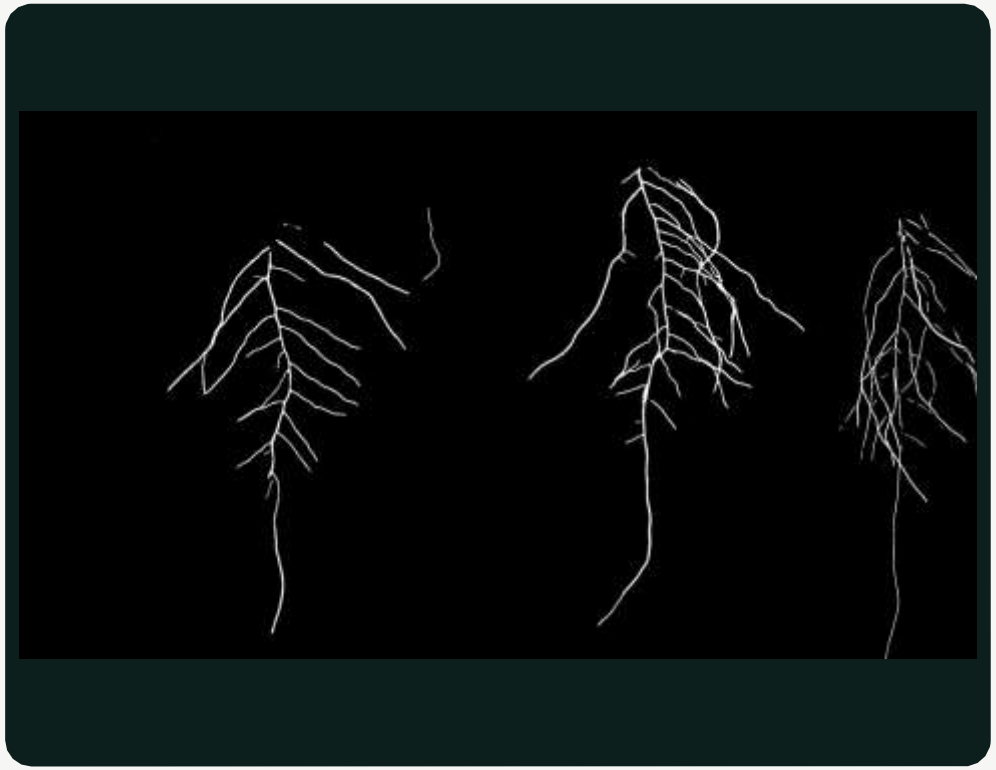


Thresholding + connected components + directional morphology + global root-to-shoot assignment

Source: corrected instant segmentation notebook (both test images)

The 512 × 512 root segmentation model

Residual U-Net · group normalization · Dice + BCE + soft-clDice

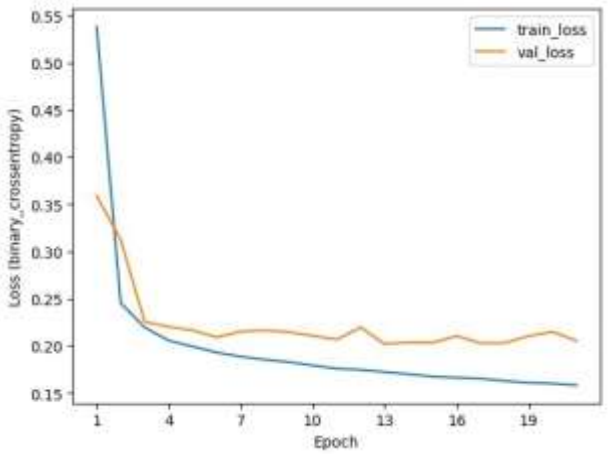


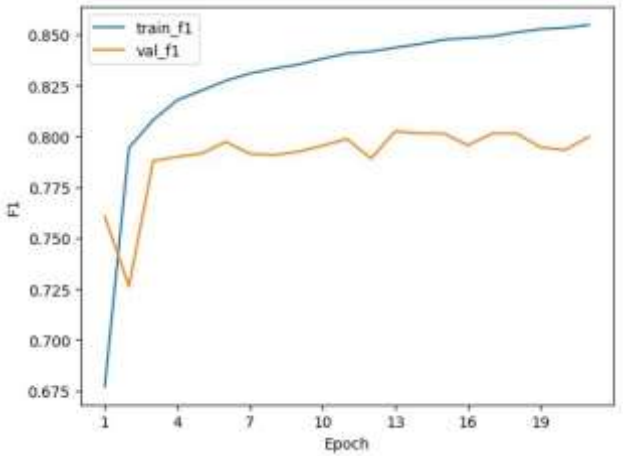
0.8024

best validation F1

Train: 9,600 patches · Y2B_23 / _24

Val: 612 patches · Y2B_25



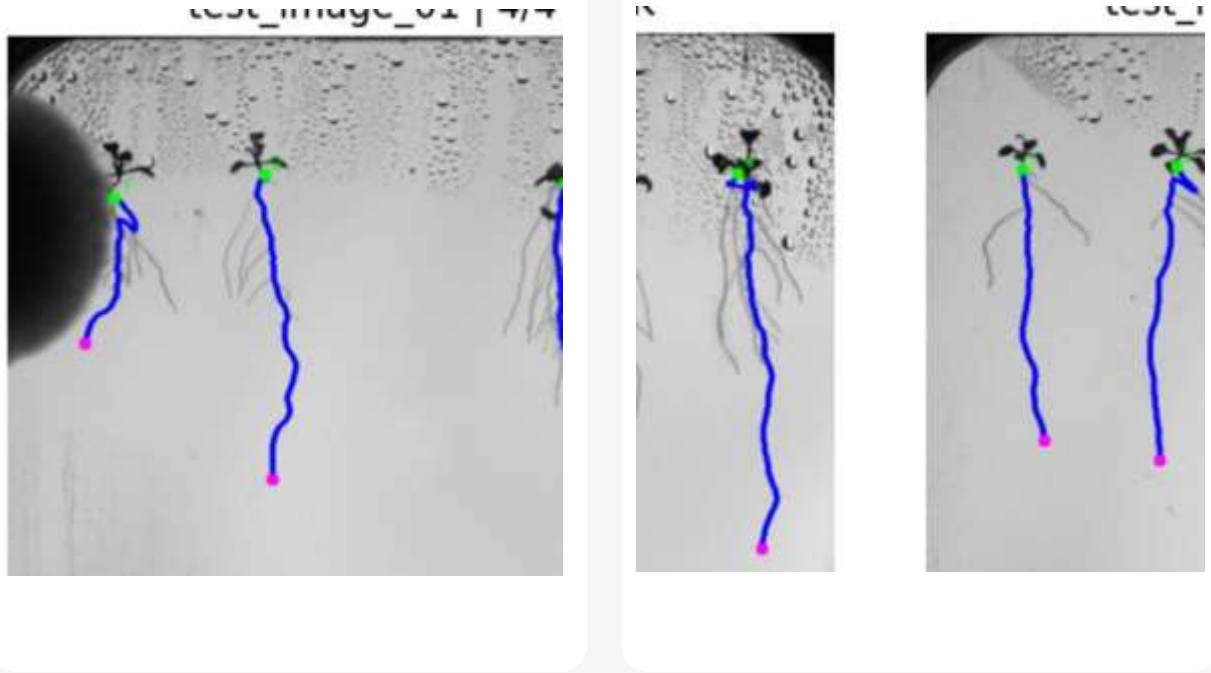


Faint or disconnected roots remain difficult, despite learned structure.

Source: Model training notebook + binary prediction image

From a pixel mask to primary root measurements

Root-start refinement V5.1 → plant assignment → traced root length



19/19
images processed

87
root measurements

11
flagged for review

76
labelled OK

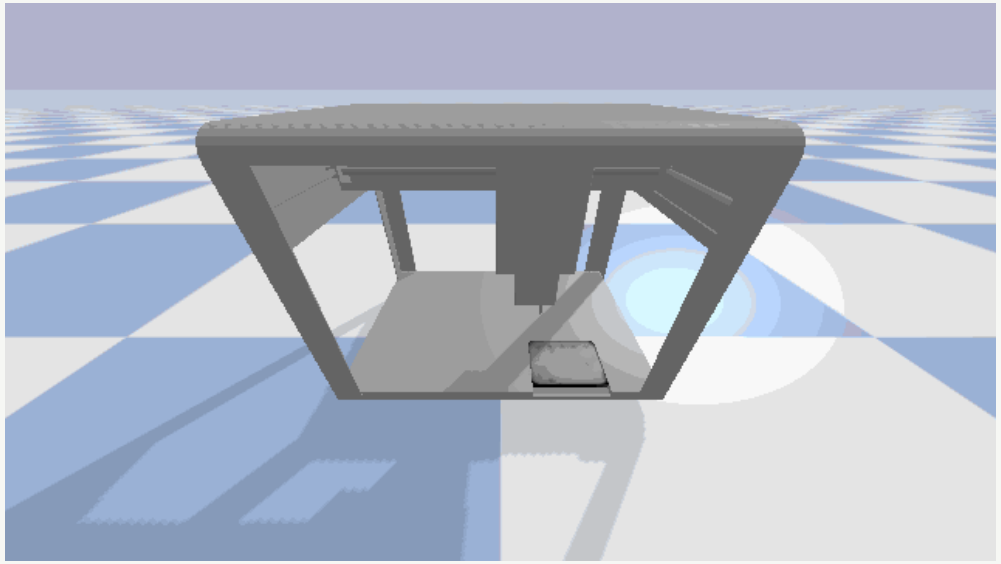
16.159%
FINAL PRIVATE KAGGLE sMAPE

Measurement quality flags are diagnostic; they do not guarantee accuracy.

Source: Task 8 notebook/README + Kaggle private screenshot

Mapping root tips into the OT-2 workspace

Reachability + image-to-robot integration



WORKFLOW

1. Root-tip pixel (x, y)
2. Assumed image-to-plate transform
3. Mapped robot target (x, y, z)
4. **Approach · hover · retreat**

Calibration caveat: simulator geometry was checked, but real photo-to-robot registration remains unverified.

Source: Eight vorner GIF + calibration checks/README

Two controllers, one positioning objective

Both target continuous 3-D pipette positioning inside the same simulation

CLASSICAL CONTROL

PI controller

Tuned proportional + integral feedback

- Independent gains for X, Y, Z
- Derivative gain set to zero

REINFORCEMENT LEARNING

Soft Actor-Critic (SAC)

Learned 12-D observation → continuous actions

87 / 100 settled on independent holdout

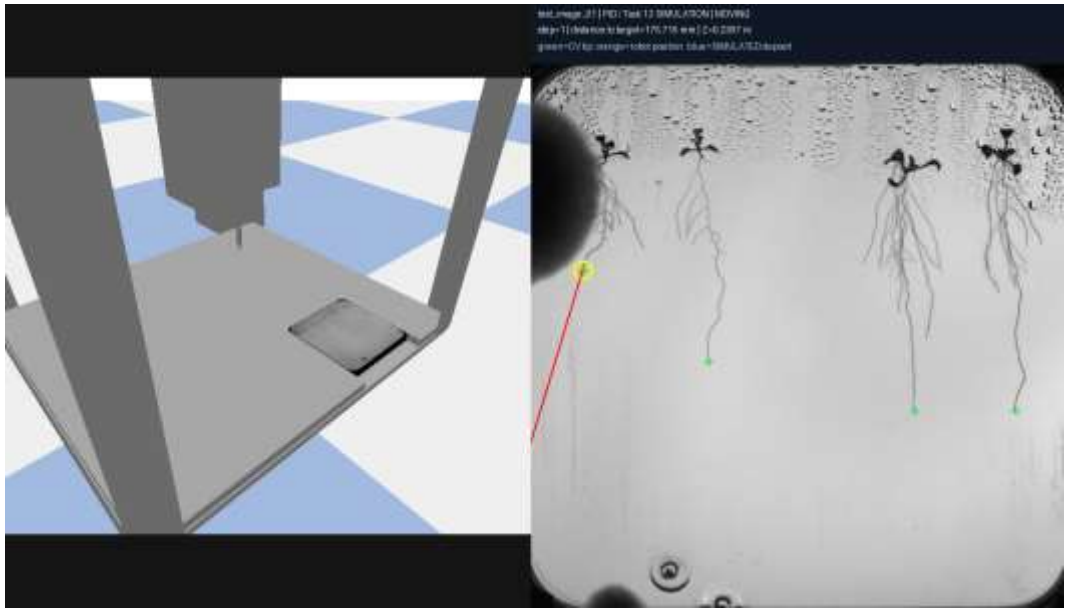
- No PID teacher at inference

Same goal: get within 1 mm, then remain there long enough to count as settled.

Autonomous route demonstrations

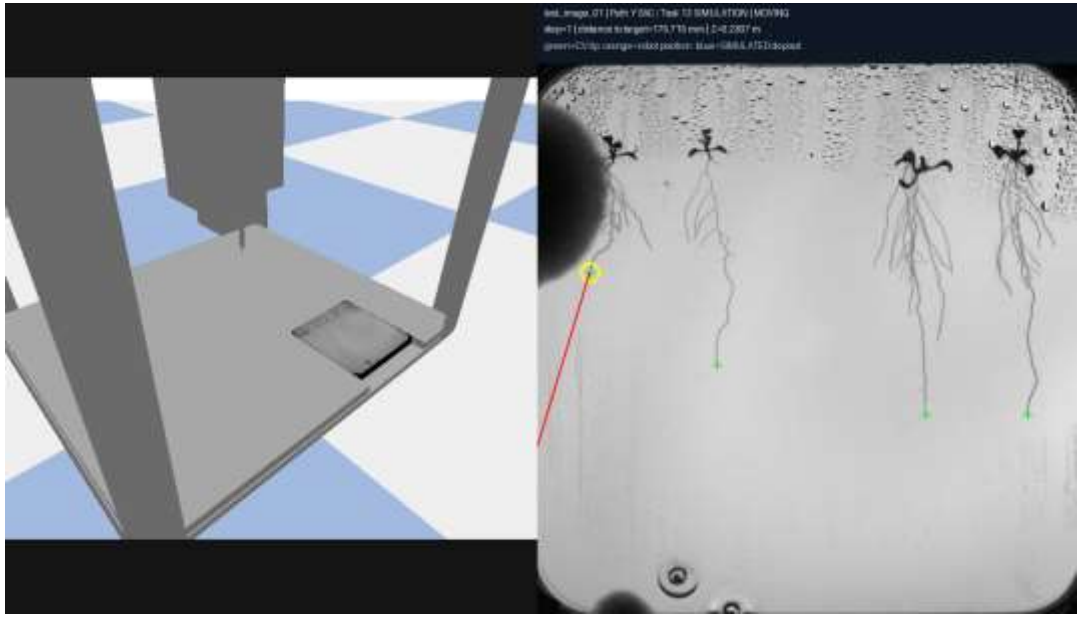
Three specimen images · 13 accepted computer-vision-derived targets

TUNED PI



13/13 simulated route markers

SAC POLICY



9/13 simulated route markers

Markers represent simulated success events, not liquid dispensing or biological inoculation.

Source: saved simulation GIFs; different protocol from slide 10

PI outperforms SAC on matched targets

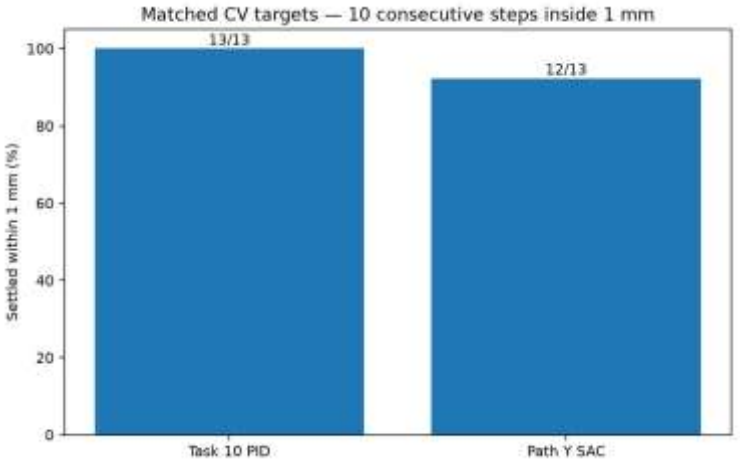
Identical 13 targets · reset pose · 600-step budget · 10-step dwell inside 1 mm

13/13
PI settled

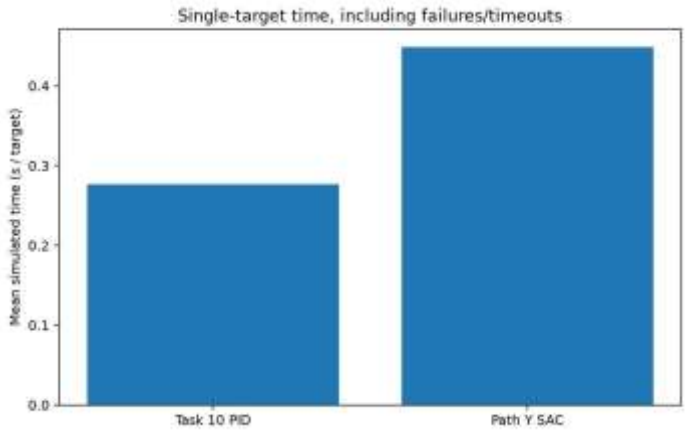
12/13
SAC settled

0.179 mm
PI mean final error

0.561 mm
SAC mean final error





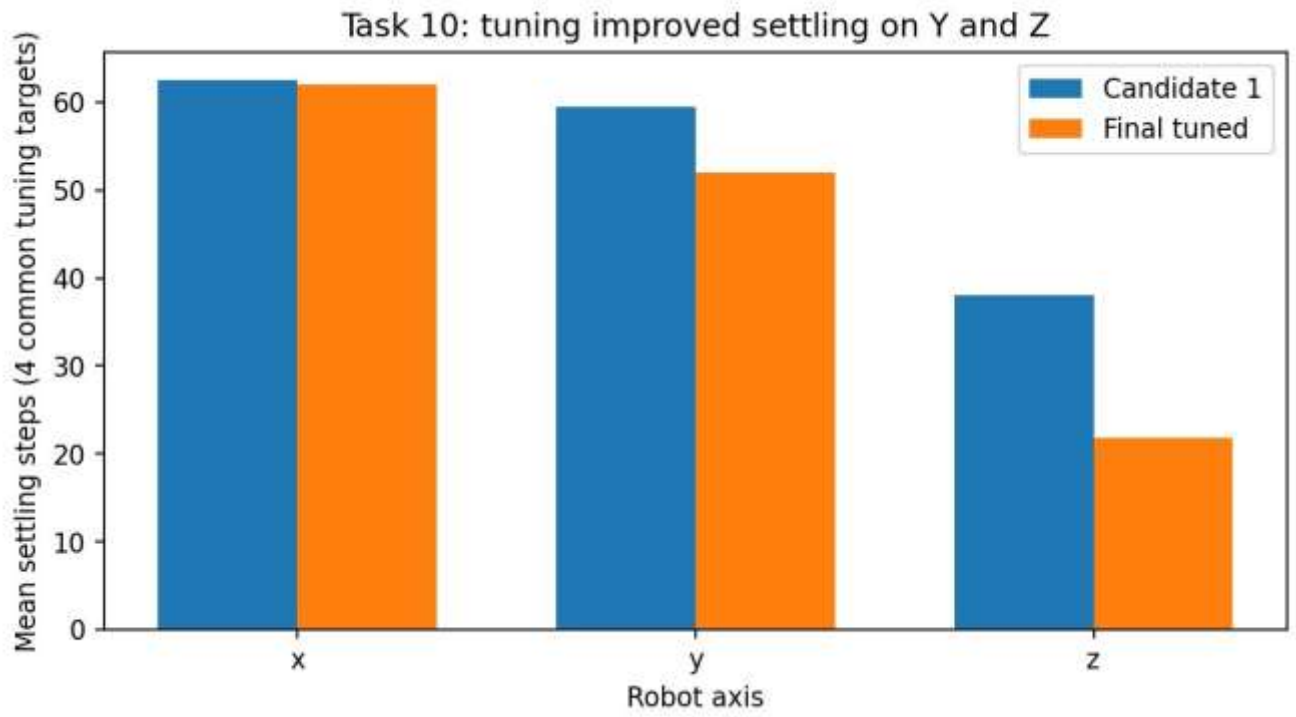


Mean simulated time per target: PI 0.277 s • SAC 0.448 s | No unsafe terminations

Source: matched_targets.csv + benchmark summary

Tuning the PI positioning controller

Error: different axes required different correction behaviour



Y 59.5 → 52

mean settling steps

Z 38 → 21.75

mean settling steps

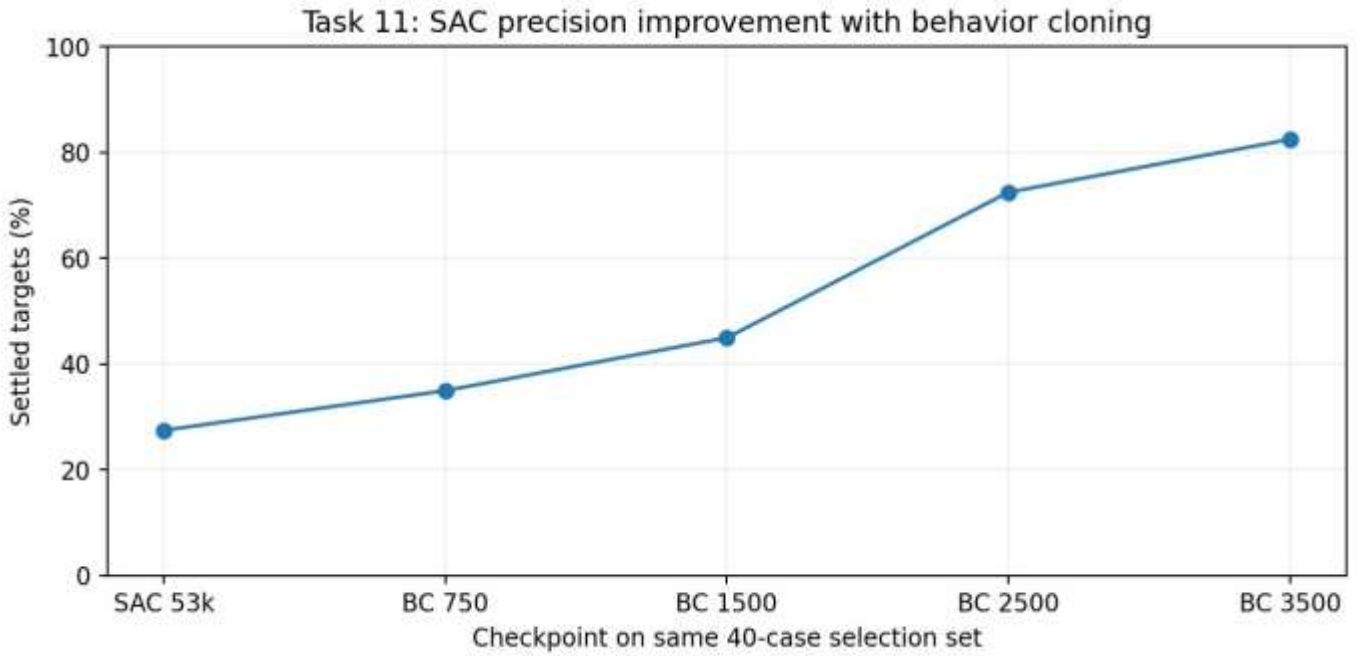
FINAL DESIGN

Separate axis gains
Kd = 0 (PI)

Source: tuning_results.csv + gains_final.json

SAC learned to approach — but not settle

Behavior cloning from successful controller trajectories improved precision



11/40 → 33/40

settled targets

1.938 → 0.717 mm

mean final error

Holdout: 87/100 settled

Additional guided RL did not beat BC.

Source: Path Y final_summary.json (same 40 selection cases)

What the results do and do not show

Assumptions define the valid scope of the system

ASSUMPTIONS

- Five seedlings in the Task 3 image layout
- Comparable HADES image appearance
- Known simulated plate geometry
- Assumed image-to-robot mapping

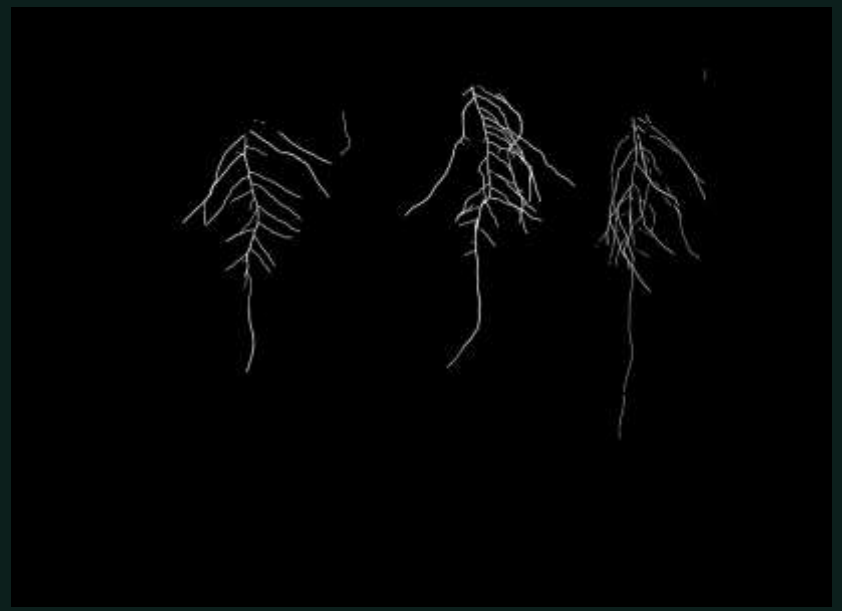
LIMITATIONS

- Faint and overlapping root structures
- Only 13 selected integrated targets
- No measured physical registration
- No real droplet / infection validation

All reported controller errors are to simulated mapped coordinates — not independently measured biological root-tip locations.

The reliable controller today is PI.

The meaningful result is a working simulated pipeline — from biological images to autonomous robot positioning.



13/13	0.179 mm	0
PI settled	PI mean error	unsafe episodes

NEXT: stronger root detection / measured calibration / real OT-2 deposition tests

For NPEC: a foundation for scalable, consistent plant–microbe experiments.